

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 36

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Photons of light absorbed by the (n-type silicon) upper layer dislodge electrons, making it negatively charged (1) • Electrons flow through an external conductor / circuit to the relatively positive lower layer; the current powers appliances (1)	2
01.2	<i>1 mark for each point.</i> • Smooth cells reflect about 30% of the light; grooved / textured surfaces (mimicking moth eyes) reflect light into the cell rather than away (1) • Multi-junction cells have several layers of different materials, each absorbing different wavelengths (1) • So a greater proportion of the available light is absorbed and converted to electricity (1)	3
01.3	<i>1 mark for each point.</i> • Solar farm: polycrystalline / thin film - lowest cost per unit electricity where land area is not limiting (efficiency per m² matters less) (1) • Satellite: multi-junction - highest efficiency where mass / area is critical and cost is less important (1) • Cost per watt = cost / efficiency: thin film 0.125, poly 0.125, mono 0.136, multi-junction 0.625 per unit (1)	3
01.4	<i>1 mark for each point.</i> • Intermittent: no output at night and low in winter (daily and seasonal cycles) - predictable (1) • Unpredictable: cloud cover, dust and smoke reduce intensity and scatter light and cannot be forecast accurately (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
02.1	<i>1 mark for each point.</i> - t-test (1) - The data are measured (continuous) values and two means are being compared (1) - The data are (assumed to be) normally distributed (1)	3
02.2	<i>1 mark.</i> Use the Mann-Whitney U test instead (compares medians) (1)	1
02.3	<i>1 mark for each point.</i> - There is a 2% probability that the difference occurred by random chance (1) - As this is less than 0.05 the difference is statistically significant / the null hypothesis is rejected (1)	2
02.4	<i>1 mark for each point.</i> - The ranges of values (mean plus or minus SD) do not overlap: woodland up to 1.6 and clearing down to 2.4 (1) - About 68% of values lie within one SD of each mean so the two data sets are clearly separated (1)	2
02.5	<i>1 mark for each point.</i> - Significance shows the difference is unlikely to be due to chance, not that a causal relationship exists (1) - Another variable (eg topography / aspect of the sites) could cause both; a controlled experiment or further evidence is needed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 420-423; AQA spec 3.7.2.3.	
03.1	<i>2 marks for each point (partial credit for a partly correct point).</i> - Orientation: south-facing windows (northern hemisphere) maximise passive solar gains; rooms needing warmth on the sunny side; fewer / smaller windows on the north side where losses are greatest; solar screens / brise soleil prevent summer overheating (2) - Low SA:V (compact shapes, terraced / joined buildings, fewer storeys of small footprint) means less external surface per unit volume through which heat is lost (2) - Earth sheltering: the ground is warmer than winter air and shields the building from wind, so conduction and wind-chill losses are reduced (2)	6
03.2	<i>1 mark for each point.</i> - Cement manufacture (heating limestone) is very energy intensive so concrete has high embodied energy (1) - Limecrete has about half the embodied energy of concrete; rammed earth, straw bales and lime mortar are lower still (1)	2
03.3	<i>1 mark for each point.</i> - Warm stale air leaving and cold fresh air entering pass through the exchanger in opposite directions (counter-current) separated by thin conducting pipes / plates with a large surface area (1) - Heat transfers from the outgoing to the incoming air so fresh air enters already warmed (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 228-232; AQA spec 3.3.4.2.2.	

Question	Marking guidance	Marks
04.1	<i>1 mark for each point.</i> - Dissolved oxygen fluctuates daily / 24-hour cycle: highest in the afternoon (about 13-13.5 mg dm⁻³ at 16 and 40 h) and lowest at night / early morning (2.5-3.5 mg dm⁻³ at 4 and 28 h) (1) - The night-time minimum is lower on the second night (2.5 vs 3.5) (1)	2
04.2	<i>1 mark for each point.</i> - Nitrates (from fertiliser runoff) and phosphates (from sewage effluent) cause eutrophication: rapid growth of algae / cyanobacteria (algal bloom) (1) - During the day photosynthesis by the dense algae releases oxygen so concentrations rise (above the normal saturation level) (1) - At night there is no photosynthesis but respiration by algae / animals / bacteria continues, using oxygen (1) - Aerobic bacterial decomposition of dead algae / shaded macrophytes uses oxygen; warm summer water holds less oxygen and decay is rapid (1)	4
04.3	<i>1 mark for each point.</i> - At night no photosynthesis replaces the oxygen used in respiration / decomposition so concentrations fall below the minimum fish can tolerate (1) - At the end of summer large amounts of dead vegetation decay rapidly in warm water causing severe deoxygenation (1)	2
04.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Cyanobacteria release toxins harmful to livestock, pets and people (1) - Floating algae shade submerged / rooted macrophytes which die, disrupting food webs (1) - Algae are an unreliable food source (die off rapidly) so food webs are poor (1) - Blocked filters / unpleasant taste and smell in water supplies (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 266-268; AQA spec 3.4.3.2.9.	
05.1	<i>1 mark.</i> The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) (1)	1
05.2	<i>Any 4 from the points below - 1 mark each, maximum 4.</i> - Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions (1) - Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition (1) - Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils (1) - pH / presence of other chemicals may increase or decrease the rate of degradation (1) - Adsorption onto clay / organic matter may protect from degradation and increase persistence (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	

Question	Marking guidance	Marks
06.1	<i>1 mark for each point.</i> • Most energy comes from non-renewable fossil fuels whose reserves are depleting; the rate of new discoveries has fallen sharply since the 1980s (1) • Use causes unacceptable environmental damage: atmospheric pollution, climate change, oil pollution, habitat loss (1) • Economic / social: depleting resources become more expensive, reducing future affluence; risk of an energy gap if new technologies are not developed in time (1)	3
06.2	<i>1 mark for each point.</i> • Coal mining: habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise (1) • Oil and gas extraction: spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO₂ / CO₂ / smoke (1) • Pipelines causing habitat damage during construction; tanker transport risks (1) • Fuel processing (refineries, coal washing) pollution; embodied energy of equipment and site development for all resources (1)	4
06.3	<i>1 mark for each point.</i> • New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) (1) • Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise (1) • By then depletion would reduce supply faster than replacements could be built, causing shortages (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.	
07.1	<i>1 mark for the correct answer.</i> • $25 + 5 = 30\%$ (1)	1
07.2	<i>2 marks for each point (partial credit for a partly correct point).</i> • UV: short wavelength; mostly absorbed in the stratosphere by monatomic, diatomic and triatomic oxygen (ozone layer) which are formed and destroyed in a dynamic equilibrium; converted to chemical energy in these reactions; little reaches the surface (2) • Visible light: transmitted through the atmosphere easily (some reflected by clouds / dust); absorbed by the Earth's surface and converted to heat; some absorbed by chlorophyll and converted to chemical energy in photosynthesis (2) • IR: emitted by the warm surface (long-wave); absorbed by greenhouse gases (CO₂, water vapour, methane) and converted to heat; the warm atmosphere re-emits IR to the surface and reduces conduction losses; some escapes to space (2)	6
07.3	<i>1 mark for each point.</i> • Greenhouse gases absorb most of the IR emitted by the surface and re-radiate it in all directions, including back down (1) • The surface is repeatedly warmed and re-emits, so the total IR exchanged between surface and atmosphere exceeds the incoming solar energy; only the net loss (70%) balances the absorbed solar energy at equilibrium (1)	2

Question	Marking guidance	Marks
07.4	<i>1 mark.</i> The atmosphere forms layers (troposphere, stratosphere, mesosphere) with different temperature gradients because energy is absorbed at different altitudes (eg UV heating the stratosphere, surface heating the troposphere) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
08.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Volume of ash is much less than the original waste (less landfill) (1) - Heat used for district heating / electricity generation (energy recovery) (1) - No sorting / complicated management needed (1) - Destroys pathogens / flammable organic wastes converted to CO₂ and water (1) - Refuse-derived fuel can replace fossil fuels (eg cement kilns) (1)	3
08.2	<i>1 mark for each point.</i> - High temperature and rapid cooling reduce the formation of toxic dioxins (from organic wastes reacting with chlorine) (1) - Scrubber dissolves hydrogen chloride (from chlorine in PVC) which is then neutralised with alkali / limestone; also removes heavy metals for storage at high pH (1) - Wet waste needs extra fuel to burn and lowers combustion temperature; composting recovers its nutrient value (1) - Recycling conserves the resource value of metals, glass and plastics that would otherwise be lost (1)	4
08.3	<i>1 mark for each point.</i> - Ash may contain concentrated heavy metals / toxic residues (1) - Must be stored in solid form at high pH / hazardous waste landfill so the metals remain insoluble (1)	2
08.4	<i>Any 1 from the points below - 1 mark each, maximum 1.</i> - Fuel (natural gas) needed to burn wet / non-flammable wastes (1) - Cost of treating effluent gases (SO₂, NO_x, HCl, smoke) (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 283-284; AQA spec 3.4.3.2.12.	
09.1	<i>1 mark for the correct answer.</i> $(150 - 125) / 150 \times 100 = 16.7\%$ (accept 17%) (1)	1
09.2	<i>1 mark for each point.</i> - A fixed fee gives no incentive to save water because the bill is the same however much is used (1) - Metering matches the charge to the volume used so households save money by using less (fewer hosepipes, low-flow appliances) (1)	2
09.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Low-water appliances: dual-flush toilets, low-flow shower heads, efficient washing machines / dishwashers, tap timers (1) - Greywater re-use: bath / shower water used to flush toilets or water gardens (1) - Rainwater collection / water butts for garden use (1) - Xeriscaping: planting drought-adapted species that need no watering (1) - Repairing leaks / dripping taps (1)	3

Question	Marking guidance	Marks
09.4	1 mark for each point. • Old infrastructure / pipes in poor condition leak (1) • Repair and good maintenance / pipe replacement / leak detection reduce wastage (1)	2
09.5	1 mark for each point. • Water is delivered directly to the plant roots / soil around each plant (1) • Less evaporation loss from spray droplets and from wetting the whole soil surface; less runoff (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 124-126; AQA spec 3.2.2.5.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • State of matter: solids deposit near the source, gases disperse widely in the atmosphere; density (lead dust, hydrogen cyanide at Bhopal) controls how far particles/gases travel • Persistence / half-life: persistent pollutants (organochlorines, CFCs, heavy metals, radioactive wastes with long half-lives) travel further, accumulate and need containment rather than dilution; non-persistent pollutants (sewage, organophosphates) can be allowed to degrade • Toxicity, specificity, mode of action (enzyme inhibition, neurotoxins) determine acceptable concentrations and which species are at risk; LD50 / MDAF / MDNF used to set doses • Solubility in water (nitrates disperse and leach - control by nitrate vulnerable zones) vs liposolubility (DDT, mercury bioaccumulate and biomagnify - control by bans / restrictions) • Reactivity: reactive pollutants may degrade quickly or form secondary pollutants (ozone, acid rain, PANs) - controlling primary pollutants (catalytic converters, FGD) prevents secondary pollutants • Adsorption onto clay / sediments immobilises metals and caesium; can be used deliberately (activated carbon filters, polymer adsorption) • Mutagenic / carcinogenic / teratogenic action means no safe dose; exposure kept As Low As Reasonably Achievable (ALARA) • Synergism (ozone + SO₂; smoke + SO₂) means pollutants must be considered together • Environmental features (temperature inversions, currents, pH, oxygen) interact with properties: Critical Pathway Analysis uses properties + environment to predict where pollutants go and to focus monitoring • Evaluation: properties of new chemicals can be predicted from similar ones (precautionary principle); understanding allows prevention rather than clean-up	9

Question	Marking guidance	Marks
10.2	1 mark. The increase in the amount / concentration of a substance within an (individual) organism over time, because intake exceeds excretion or breakdown (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240; AQA spec 3.4.1.	
11.1	Discuss the extent to which energy conservation technologies could reduce the need for new energy supplies in a developed country. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) - Scale of the problem: how energy is used (transport, buildings, industry, water); links between affluence and per capita use - Transport: bulk / integrated transport, ATM, vehicle design (aerodynamics, mass, tyres, KERS, combustion efficiency), low embodied energy, design for end of life, driver behaviour, electric vehicles - Buildings: orientation and passive solar, SA:V, thermal mass, low embodied energy materials, earth sheltering, glazing (low-e, inert gas, vacuum), insulation optimum, heat exchangers, automatic ventilation, occupancy sensors, thermostats, low-energy appliances, behaviour - Industry: heat recovery and exchangers, insulation and high-volume storage, CHP, integrated manufacture, recycling and mass reduction, high-voltage grid, peak shaving, ICT management - Limits: rebound effect, cost and payback, existing building stock, embodied energy of replacements, behaviour - Evaluation with quantitative examples; comparison with the cost of new supply; conclusion that conservation is the cheapest first step but cannot alone meet demand growth Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 170-174, 225-235; AQA spec 3.3.4.2, 3.3.1.	25

Question	Marking guidance	Marks
11.2	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25